

AegisPay — Build Errors & How We Solved Them

Sif Error aur uske Solution. Har naye error neeche add karo (points 4 ko chhuro)

PDF dobara banane ke liye: `python pdf/build_error_guide_pdf.py`

1. Docker app: ModuleNotFoundError: No module named 'api'

- Error: app container crash loop — No module named 'api'.
- Problem: Build context api/ tha, lekin package repo root par hai; COPY . . se modules /app mein flat ho gaye aur import api fail.
- Fix: Context = repo root, aur COPY api ./api kar phir pip install.

2. Migration: connection to server at "db" ... Connection refused

- Error: db_migrate started before Postgres ready; connection refused, exit 2.
- Problem: db par koi healthcheck nahi tha; race condition.
- Fix: db healthcheck add karo + depends_on: db: condition: service_healthy.

3. Migration: fe_sendauth: no password supplied

- Error: psql se no password supplied.
- Problem: TCP par password required tha, container mein PGPASSWORD nahi tha.
- Fix: PGPASSWORD: aegispay env mein set karo (+ ON_ERROR_STOP=1).

4. Localstack: exit code 55 ... License activation failed

- Error: LocalStack quit, "No credentials / license".
- Problem: localstack:latest ab auth/license maangta hai.
- Fix: Free OSS image pin karo — localstack/localstack:3.5.0.

5. Docker build: pip ... ReadTimeoutError ... pythonhosted

- Error: Image build mein pip download timeout.
- Problem: Default pip timeout chota tha, network slow.
- Fix: Dockerfile mein PIP_DEFAULT_TIMEOUT=120 + --timeout 120.

6. CI test: '_IncludedRouter' object has no attribute 'path'

- Error: Smoke test fail — 1 failed, 40 passed.
- Problem: Test app.routes par route.path padh raha tha; FastAPI 0.141.1 (latest) internal _IncludedRouter deta hai. Local had 0.135.2 -> local pass, CI fail.
- Fix: app.openapi()["paths"] use karo (stable). FastAPI pin: >=0.115, <0.142.

7. PDF renderer: saw instead of expected </i>

- Error: ReportLab parse error while rendering inline code/emphasis.
- Fix: Code spans ko placeholder se replace karo (pehle), phir bold/italic, phir restore.

8. PDF build: ModuleNotFoundError: No module named 'pdf'

- Error: python pdf/build_x.py chalane par pdf module nahi mila.
- Fix: Script ke top par repo root sys.path mein add karo:
`sys.path.insert(0, str(Path(__file__).resolve().parents[1]))`.

9. CI gitleaks: failed to scan Git repository — "stderr is not empty"

- Error: CI test job fail — gitleaks secret scan abort.

- Problem: gitleaks HEAD ko uske parent commit se diff karta hai, par PR default shallow checkout (depth 1) se aata hai jisme parent commit nahi hota -> ambiguous argument '<sha>^...<sha>': unknown revision.
- Fix: checkout mein fetch-depth: 0 (full history) + gitleaks step ko continue-on-error: true (report-only, koi real leak nahi mila).

10. CI integration: password authentication failed for user "aegispay_app"

- Error: Integration tests fail — InvalidPasswordError.
- Problem: CI ka bare Postgres service db/init.sql nahi chalata (wo sirf local compose endpoint mein hota hai), isliye aegispay_app role banti hi nahi -> migration ke grants + tests connect fail.
- Fix: Integration workflow mein migration se pehle psql -f api/db/init.sql chalao (role create + grants).

Naya error yahan add karo (template)

```
## N. <short error title>
- **Error:** <paste error>
- **Problem:** <why it happened, 1-2 lines>
- **Fix:** <what we changed>
```